30

Kolsky

CHERRY PIE LAYOUT

copy of old 1955 study

FOR STRETCH

FOUR MEMORIES

$i \rightarrow$	1	2	3	4
F	1 ?	2	3	4	1
r^n	2	3	4	1	2
z^n	3	4	1	2	3
u	4	1	2	3	4
w	1	2	3	4	1
$z^{n+1/2}$	2	3	4	1	2
$r^{n+1/2}$	3	4	1	2	3
ρ^n	4	1	2	3	4
$\rho^{n+1/2}$	1	2	3	4	1
E	2	3	4	1	2
CONST.	3	4	1	2	3
i, L_2	4	1	2	3	4
$L_3 L_4$	1	2	3	4	1

CHERRY PIE - FIRST PASS — FOUR MEMORIES

INSTR #	INDEX	OPER CODE	TRANSFER OPERATION	ARITH. OPER.	REMARKS
1		45	LOAD CA		$i = 1$ SET INDEX
2	1	2	$MM_2 \rightarrow L$		P_1
3		45	LOAD CA		$i = 3$ SET INDEX
4	1	9	$MM_4 \rightarrow L$	F-	P_{13} SUB
5		6	$L \rightarrow FM$		STORE P_{13} $(P_1 - P_3)$ STORED
6		45	LOAD CA		$i = 2$ INDEX
7	1	2	$MM_1 \rightarrow L$		ζ_2
8		45	LOAD CA		$i = 4$ INDEX
9	1	9	$MM_3 \rightarrow L$	F-	ζ_{24} SUB
10		6	$L \rightarrow FM$		STORE ζ_{24} $(\zeta_2 - \zeta_4)$ STORED
11		10	$FM \rightarrow L$	FX	$P_{13}\zeta_{24}$ MPY
12		6	$L \rightarrow FM$		STORE $P_{13}\zeta_{24}$
13		45	LOAD CA		$i = 2$ INDEX
14	1	2	$MM_3 \rightarrow L$		P_2
15		45	LOAD CA		$i = 4$ INDEX
16	1	9	$MM_1 \rightarrow L$	F-	P_{24} $P_2 - P_4$
17		6	$L \rightarrow FM$		STORE P_{24}
18		45	LOAD CA		$i = 1$
19	1	2	$MM_4 \rightarrow L$		ζ_1
20		45	LOAD CA		$i = 3$
21	1	9	$MM_2 \rightarrow L$	F-	ζ_{13} $\zeta_1 - \zeta_3$
22		6	$L \rightarrow FM$		STORE ζ_{13}
23		10	$FM \rightarrow L$	FX	$P_{24}\zeta_{13}$
24		9	$FM \rightarrow L$	F-	$P_{13}\zeta_{24} - P_{24}\zeta_{13}$
25		6	$L \rightarrow FM$		STORE $(P_{13}\zeta_{24} - P_{24}\zeta_{13})$
26		45	LOAD CA		$i = 1$
27	1	2	$MM_3 \rightarrow L$		r_1
28		45	LOAD CA		$i = 3$
29	1	9	$MM_1 \rightarrow L$	F-	r_{13}

CHERRY PIE - FIRST PASS — FOUR MEMORIES

INSTR. #	INDEX	OPER. CODE	TRANSFER OPERATION	ARITH OPER.	REMARKS
30		6	L → FM		STORE r_{13}
31		10	FM → L	Fx	$P_{24}\, r_{13}$
32		6	L → FM		STORE $P_{24}\, r_{13}$
33		45	LOAD CA		$i = 2$
34	1	2	MM_4 → L		r_2
35		45	LOAD CA		$i = 4$
36	1	9	MM_2 → K	F−	r_{24}
37		6	L → FM		STORE r_{24}
38		10	FM → L	Fx	$P_{13}\, r_{24}$
39		9	FM → L	F−	$P_{24}\, r_{13} - P_{13}\, r_{24}$
40		6	L → FM		STORE $(P_{24}\, r_{13} - P_{13}\, r_{24})$
41		2	FM → L		
42		10	FM → L	Fx	$r_{13}\, z_{24}$
43		6	L → FM		STORE $r_{13}\, z_{24}$
44		2	FM → L		
45		10	FM → L	Fx	$r_{24}\, z_{13}$
46		9	FM → L	F−	$r_{13}\, z_{24} - r_{24}\, z_{13} = \Delta$
47		6	L → FM		STORE Δ
48		2	FM → L		
49		11	FM → L	F÷	$\frac{\partial P}{\partial z}$
50		6	L → FM		STORE $\frac{\partial P}{\partial z}$
51		2	FM → L		
52		11	FM → L	F÷	$\frac{\partial P}{\partial r}$
53		10	FM → L	Fx	$\frac{\partial P}{\partial r})\Delta t$
54		45	LOAD CA		$i = 0$
55	1	1	MM_4 → K		ρ^n
56		5	K → FM		STORE ρ^n
57		11	FM → L	F÷	$\frac{1}{\rho}\frac{\partial P}{\partial r})\Delta t$
58	1	9	MM_4 → L	F−	$u^{n+1/2}$
59	1	6	L → MM_4		STORE $u^{n+1/2}$

CHERRY PIE - FIRST PASS FOUR MEMORIES

INSTR #	INDEX	OPER. CODE	TRANSFER OPERATION	ARITH. OPER.	REMARKS
60		10	FM → L	Fx	$u^{n+1/2}\Delta t$
61	1	1	MM_2 → K		r^n
62		5	K → FM		STORE r^n
63		8	FM → L	F+	
64	1	6	L → MM_2		STORE r^{n+1}
65		8	FM → L	F+	$r^n + r^{n+1}$
66		10	FM → L	Fx	$\frac{1}{2}(r^n + r^{n+1}) = r^{n+\frac{1}{2}}$
67	1	6	L → MM_3		STORE $r^{n+\frac{1}{2}}$
68		2	FM → L		
69		10	FM → L	Fx	$\frac{\partial p}{\partial z})\Delta t$
70		11	FM → L	F÷	$\frac{1}{\rho}\frac{\partial p}{\partial z})\Delta t$
71	1	9	MM_1 →	F−	$w^{n+\frac{1}{2}}$
72	1	6	L → MM_1		STORE $w^{n+\frac{1}{2}}$
73		10	FM → L	Fx	$w\Delta t$
74	1	1	MM_3 → K		
75		5	K → FM		STORE z^n
76		8	FM → L	F+	z^{n+1}
77	1	6	L → MM_3		STORE z^{n+1}
78		8	FM → L	F+	$z^n + z^{n+1}$
79		10	FM → L	Fx	$\frac{1}{2}(z^n + z^{n+1}) = z^{n+\frac{1}{2}}$
80	1	6	L → MM_2		STORE $z^{n+1/2}$
81		46	C ADD	+	
82		79	STORE CA		
83	1	1	MM_4 → K		
84		5	K → FM		
85	1	1	MM_1 → K		
86		5	K → FM		
87		47	C SUBTRACT		
88		52	TR +		RETURN
89		51	TR		EXIT

TIME CHART - CHERRY PIE

ENGRAY NO. 5510 5X5 TO THE INCH
PRINTED IN U.S.A.

FAST MEMORY 1

FAST MEMORY 2

CONTROL ADDER

TRANSISTOR MEMORY

LARGE MEMORY 1

LARGE MEMORY 2

LARGE MEMORY 3

LARGE MEMORY 4

ARITHMETIC UNIT

69 71 73 75 77 79 81 83 85 87

70 72 74 76 78 80 82 84 86 88

71 72 74 77 80 81 82 83 85 87

68 69 70 73 75 76 78 79 82 64 66

71 72 85

64 80

67 74 77

83

66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86

16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

TIME CHART - CHERRY PIE

copy of 155 chart

FAST MEMORY 1

FAST MEMORY 2

CONTROL ADDER

TRANSISTOR MEMORY

LARGE MEMORY 1

LARGE MEMORY 2

LARGE MEMORY 3

LARGE MEMORY 4

ARITHMETIC UNIT

FAST MEMORY 1

FAST MEMORY 2

CONTROL ADDER

TRANSISTOR MEMORY

LARGE MEMORY 1

LARGE MEMORY 2

LARGE MEMORY 3

LARGE MEMORY 4

ARITHMETIC UNIT

MS 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15